Functional Requirement Workspace

Each Functional Requirement should be represented using a Use Case

Use case Template:

Number	< use case number >	
Name	< use case name - a short active verb phrase >	
Summary	< a brief summary of the use case >	
Priority	< how critical this use case is to the customer (1 to 5, 5 being most critical >	
Preconditions	< conditions that must be true before the use case trigger >	
Postconditions	< conditions that will be true after the use case completes >	
Primary Actor	< a role name for the primary actor >	
Secondary Actors	< other systems that are relied upon to accomplish the use case >	
Trigger	< the action that starts the use case >	
Main Scenario	**Step**	**Action**
	1	< steps of the use case from trigger to goal delivery >
	2	< … >

	3	< … >
Extensions	**Step**	**Branching Action**
	1a	< condition causing branching > **:** < action or name of sub use case >
Open Issues	< list of issues awaiting decisions that affect the use case >	

Functional Requirements:

Note - “This/The System shall…” // Important formatting

1. DICOM File Import
    a. The System shall allow the user to select and upload a DICOM file(or folder of DICOM files) from their local machine
    b. The system shall validate the DICOM format before processing
    c. The system shall read and parse the DICOM file to extract relevant CT scan image data and metadata(or - information?)
2. Anatomical Region Selection
    a. The system shall allow the user to select an anatomical target for visualization and output - bone, skin, muscle
    b. The system shall filter or threshold CT data based on the selected tissue type
    c. The system shall update any visualization or preview accordingly when a different region is selected - going from bone to muscle updates automatically
3. 3D Visualization
    a. The system shall generate and display a 3d preview of the reconstructed anatomy from the CT data within the web browser
    b. The visualization shall allow the user to rotate, zoom, and inspect the 3D model interactively
    c. The visualization shall update dynamically based on user selection(tissue type or output format)
4. Polygonal Model Generation(STL conversion)
    a. The system shall apply a Marching Cubes (or similar) algorithm to extract a polygonal surface representation of the selected anatomy
    b. The system shall generate a 3D mesh from the processed CT scan data
    c. The system shall export the resulting 3D model as a .STL file suitable for 3D printing or slicing
5. G-Code Generation
    a. The system shall allow the user to select an output type: .stl or .gcode

    b. The system shall generate G-code output that reproduces the Xray attenuation properties of the tissue - mimicking density or opacity
    c. The system shall write the generated G-code to a user-specified output location on the local machine
6. File Output
    a. The system shall prompt the user to enter or confirm an output file name
    b. The system shall save the resulting .stl or .gcode to a local folder chosen by the user
    c. The system shall notify the user when the file generation is complete or if there is an error
7. User Interface
    a. The system shall provide an intuitive and responsive graphical user interface that allows users to:
        i. Select input files(DICOM)
        ii. Choose anatomy type(bone, skin, muscle)
        iii. Choose output type(.stl or .gcode)
        iv. Visualize results in 3D view
        v. Choose output file name and location
8. Client-Side Execution
    a. The system shall run entirely within the client's browser with no external data transfer
    b. The system shall process DICOM data locally, ensuring that medical data never leaves the user's machine
9. Cross-Browser and Cross-Platform Compatibility
    a. The system shall operate correctly on Chrome, Safari, and Firefox
    b. The system shall function across MacOS, Windows, and Linux operating systems
10. Automated Testing and Verification
    a. The system shall include an automated test suite to verify correct operation across supported browsers and OS platforms
    b. The system shall use a web testing framework (e.g., Selenium) to validate core functionality
11. ???

Use Case Formatting:

Use Case -1: Import DICOM

Number	1
Name	Import DICOM File

Summary	The user selects and uploads a DICOM file or folder of DICOM files containing CT scan data for processing	
Priority	5	
Preconditions	1. The web app is loaded in the user's browser 2. The user has valid DICOM data stored locally	
Postconditions	1. The DICOM file is validated and parsed 2. CT image data and metadata are ready for processing	
Primary Actor	User	
Secondary Actors	Web browser file Input/Output API	
Trigger	User selects "Upload DICOM" or equivalent option in the interface	
Main Scenario	**Step**	**Action**
	1	The user opens the web app
	2	The system prompts the user to select a DICOM file or folder
	3	The user selects one or more DICOM(.dcm) files from their local machine
	4	The system validates the selected files to make sure they're in the DICOM format
	5	The system parses the DICOM data and extracts image slices and metadata

	6	The system notifies the user that the import was successful or reports an error if validation fails
Extensions	**Step**	**Branching Action**
	4a	If the selected file is not a valid DICOM format **:** The system displays an error and prompts the user to try again
	5a	If the file is corrupted: The system alerts the user and terminates the import process
Open Issues	None	
Verification Tests	1. Load valid DICOM: Try with different sizes and number of files. 2. Invalid file: error shown	

Use Case -2: Select Anatomical Region

Number	2
Name	Select Anatomical Region
Summary	The user chooses which anatomical region(bone, skin, or muscle) to visualize and output
Priority	4
Preconditions	User case 1 complete
Postconditions	Segmentation parameters for the target are active.

Primary Actor	End user	
Secondary Actors	• DICOM Processing Module • Browser Interface	
Trigger	User selects tissue option	
Main Scenario	**Step**	**Action**
	1	The system displays tissue options
	2	User selects one.
	3	System applies preset thresholds/range
Extensions	**Step**	**Branching Action**
	1a	Custom thresholds parameterized by user
Open Issues	None	
Verification Tests	1. Selecting “bone” applies bone preset 2. Try all the other presets	

Use Case -3: Preview Output

Number	3
Name	Visualize and Inspect 3D Anatomy
Summary	App previews the selected anatomy like surface or slices

Priority	4	
Preconditions	User case 1 and 2 completed	
Postconditions	Preview is rendered user can accept or adjust	
Primary Actor	End user	
Secondary Actors	• Rendering Engine: WebGL/WebGPU - generates real-time previews • Segmentation Engine - provides voxel data for rendering • UI Controller - manages preview controls	
Trigger	User opens Preview	
Main Scenario	**Step**	**Action**
	1	System generates isosurface/volume preview
	2	User rotates and zooms
	3	toggles between different level of threshold
	4	User accepts preview
Extensions	**Step**	**Branching Action**
	1a	Takes a long time or a large input file: Lower resolution preview
Open Issues	None	

Verification Tests	1. Preview appears within N seconds on each supported browser 2. Threshold change updates preview

Use Case -4: Generate STL

Number	4	
Name	Create Polygonal Mesh (.stl export)	
Summary	Produce an STL mesh of the selected anatomy via Marching Cubes and exports it as an .stl file	
Priority	5	
Preconditions	User case 1 and 2 completed	
Postconditions	STL mesh is created in memory and an .stl file gets downloaded	
Primary Actor	End User	
Secondary Actors	● Marching Cubes Algorithm Module: performs isosurface extraction ● Mesh Post-Processing Engine: repairs or smooths geometry ● System Memory: temporarily holds output mesh data	
Trigger	User clicks “Generate STL”	
Main Scenario	**Step**	**Action**
	1	System runs Marching Cubes with current thresholds.
	2	System post-processes - smoothing, missing spots

	3	Mesh statistics displayed
Open Issues	None	
Verification Tests	1. Output .stl is readable by standard slicers 2. Mesh bounds match CT bounds within tolerance	

Use Case -5: Generate G-code

Number	5	
Name	Generate 3D Print Instructions(.gcode export)	
Summary	Produce G-code that mimics tissue x-ray characteristics when printed	
Priority	4	
Preconditions	User case 1 and 2 completed	
Postconditions	G-code held in memory and .gcode gets downloaded	
Primary Actor	End User	
Secondary Actors	Depends on the algorithm we get from our client	
Trigger	User clicks “Generate G-code”	
Main Scenario	**Step**	**Action**

	1	Depends on the algorithm we get from our client
Extensions	**Step**	**Branching Action**
	1a	Optional hardware profile templates like: printer nozzle, materials
Open Issues	None	
Verification Tests	1. G-code loads in a standard host 2. Same output across all browsers	